

Lagrange Interpolation as Normalized Pandrosion Quotients

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Abstract

The Lagrange interpolation basis polynomials are exactly the *normalized Pandrosion quotients*: $L_j(z) = Q_j(z)/P'(\alpha_j)$, where $Q_j(z) = P(z)/(z - \alpha_j)$ is the Pandrosion quotient at the j -th node. The proof is one line: $P = (z - \alpha_j) Q_j$ gives $P'(\alpha_j) = Q_j(\alpha_j)$, so $L_j(\alpha_j) = 1$. The interpolation formula $f(z) = \sum f(\alpha_j) Q_j(z)/P'(\alpha_j)$ is the Pandrosion decomposition of f along the fiber $\{\alpha_1, \dots, \alpha_n\}$.

1 The identity

Theorem 1.1. *For distinct nodes $\alpha_1, \dots, \alpha_n$ and $P(z) = \prod (z - \alpha_j)$:*

$$\boxed{L_j(z) = \frac{Q_j(z)}{P'(\alpha_j)} = \frac{\prod_{k \neq j} (z - \alpha_k)}{\prod_{k \neq j} (\alpha_j - \alpha_k)}, \quad L_j(\alpha_k) = \delta_{jk}.} \quad (1)$$

Proof. Since $P(z) = (z - \alpha_j) Q_j(z)$, differentiating and evaluating at $z = \alpha_j$:

$$P'(\alpha_j) = Q_j(\alpha_j) + (\alpha_j - \alpha_j) Q_j'(\alpha_j) = Q_j(\alpha_j).$$

Therefore $L_j(\alpha_j) = Q_j(\alpha_j)/P'(\alpha_j) = 1$. For $k \neq j$: $Q_j(\alpha_k) = 0$ (since α_k is a root of Q_j), so $L_j(\alpha_k) = 0$. $\square$

Verified: the matrix $[L_j(\alpha_k)]$ equals the 4×4 identity matrix at machine precision.

2 Interpolation as Pandrosion decomposition

Corollary 2.1. *For any function f :*

$$I_n f(z) = \sum_{j=1}^n f(\alpha_j) \frac{Q_j(z)}{P'(\alpha_j)}, \quad (2)$$

the unique polynomial of degree $< n$ passing through $(\alpha_j, f(\alpha_j))$.

This is a *decomposition along the Pandrosion fiber*: the value of f at each root α_j is weighted by the corresponding quotient Q_j .

3 Partition of unity

Proposition 3.1. $\sum_{j=1}^n L_j(z) = 1$ for all z (*partition of unity*).

Proof. Interpolate $f \equiv 1$: $I_n(1) = \sum L_j = 1$. $\square$

Equivalently, $\sum Q_j(z)/P'(\alpha_j) = 1$, which is the Pandrosion identity $\sum Q_j = P'$ (Paper 1) combined with $Q_j(\alpha_j) = P'(\alpha_j)$.

Verified: $\max |\sum L_j - 1| < 10^{-14}$ for three sets of nodes.

4 Connection to the Pandrosion field

The partial fraction decomposition of $F_P = P'/P$ can be written:

$$F_P(z) = \sum_{j=1}^n \frac{1}{z - \alpha_j} = \sum_{j=1}^n \frac{P'(\alpha_j)^{-1} Q_j(z)}{Q_j(z) (z - \alpha_j)/P(z)} = \dots \quad (3)$$

More directly: the *residue* of $f(z)/P(z)$ at $z = \alpha_j$ is $f(\alpha_j)/P'(\alpha_j)$, and the Lagrange formula is precisely the statement that:

$$\frac{1}{2\pi i} \oint_{\gamma} \frac{f(\zeta) P(z)}{P(\zeta) (z - \zeta)} d\zeta = \sum_{j=1}^n \frac{f(\alpha_j) Q_j(z)}{P'(\alpha_j)} = I_n f(z). \quad (4)$$

This is the Cauchy integral version of Pandrosion interpolation.

References

- [1] J.-L. Lagrange, *Leçons sur le calcul des fonctions*, 1795.
- [2] I. Besevic, *The Pandrosion quotient*, Pandrosion Dynamics **1** (2026).
- [3] I. Besevic, *Vieta–Newton unified*, Pandrosion Dynamics **90** (2026).